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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.17947011743025743, median 0.14712342382429333, std: 0.1307055011491053
Reprojection error (cam1):     mean 0.18811547601045436, median 0.15507724601598044, std: 0.13146798738597235
Gyroscope error (imu0):        mean 0.25047896616844373, median 0.2431050408432199, std: 0.11516463530973307
Accelerometer error (imu0):    mean 0.12355054173030634, median 0.10768607515493951, std: 0.08796434377512219

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.17947011743025743, median 0.14712342382429333, std:
0.1307055011491053
Reprojection error (cam1) [px]:     mean 0.18811547601045436, median 0.1550772460159804, std:
0.13146798738597232
Gyroscope error (imu0) [rad/s]:     mean 0.0015294956609381806, median 0.0014844683799591165, std:
0.0007032279504112547
Accelerometer error (imu0) [m/s^2]: mean 0.01687219912932995, median 0.014705729963015234, std:
0.012012508433143047

Transformation (cam0):
------------------------
T_ci:  (imu0 to cam0):
[[ 0.03068482 -0.99952813  0.00139887  0.01877251]
[-0.01883952 -0.00197764 -0.99982056  0.03163655]
[ 0.99935155  0.03065296 -0.01889132 -0.09139607]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.03068482 -0.01883952  0.99935155  0.09135679]
[-0.99952813 -0.00197764  0.03065296  0.02162777]
[ 0.00139887 -0.99982056 -0.01889132  0.02987802]
[ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.011808052681018309


Transformation (cam1):
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T_ci:  (imu0 to cam1):
[[ 0.03017694 -0.99954399  0.00107815 -0.03121645]
[-0.02101459 -0.00171285 -0.9997777   0.03181823]
[ 0.99932364  0.03014758 -0.02105669 -0.09108664]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.03017694 -0.02101459  0.99932364  0.0926357 ]
[-0.99954399 -0.00171285  0.03014758 -0.02840167]
[ 0.00107815 -0.9997777  -0.02105669  0.02992683]
[ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.011772415116308976

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99999982  0.00033026 -0.00050198 -0.05004528]
[-0.00033135  0.9999976  -0.00216635 -0.00001002]
[ 0.00050126  0.00216651  0.99999753  0.00023126]
[ 0.          0.          0.          1.        ]]
baseline norm:  0.050045814036265006 [m]


Gravity vector in target coords: [m/s^2]
[-0.17459986 -9.80497503 -0.02006035]


Calibration configuration
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cam0
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```

Camera model: pinhole
Focal length: [387.26496788634023, 388.0178836701606]
Principal point: [320.53538193017073, 238.7961117373796]
Distortion model: radtan
Distortion coefficients: [-0.0048964025986568405, -0.0032256738628326904, 6.867803461190135e-05, 0.0003187277311499132]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [388.18861658777746, 388.9712538925426]
Principal point: [321.2136971603922, 238.95658293993972]
Distortion model: radtan
Distortion coefficients: [-0.002870215094047284, -0.00471202846643744, 5.440321557997829e-05, -9.458602566352592e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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IMU0:

Model: calibrated
Update rate: 193.393

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Accelerometer:
Noise density: 0.00981989090996197
Noise density (discrete): 0.13656111007719915
Random walk: 0.0005601293637217548
Gyroscope:
Noise density: 0.00043909309937990626
Noise density (discrete): 0.006106283830274256
Random walk: 1.6907369481278766e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```
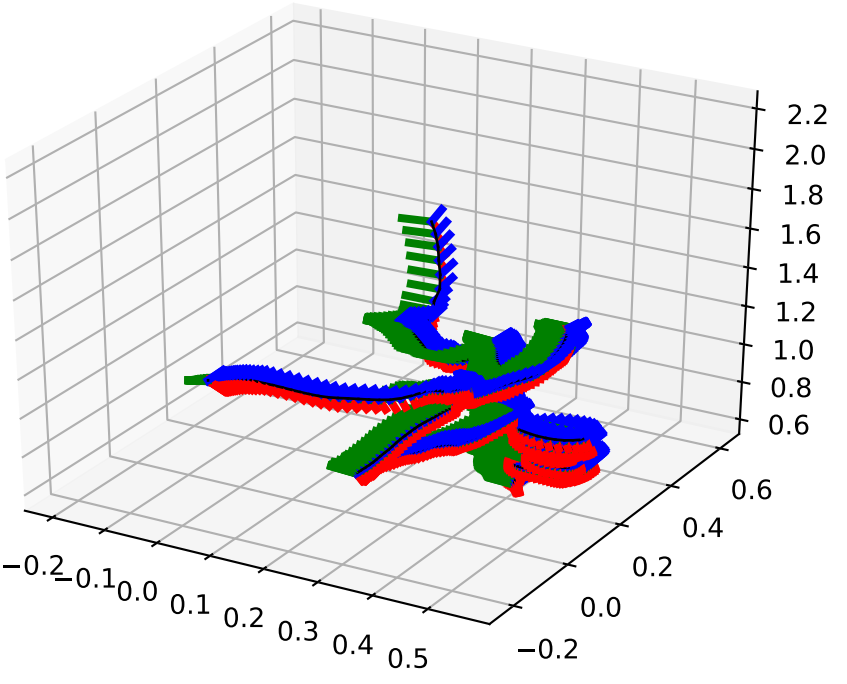
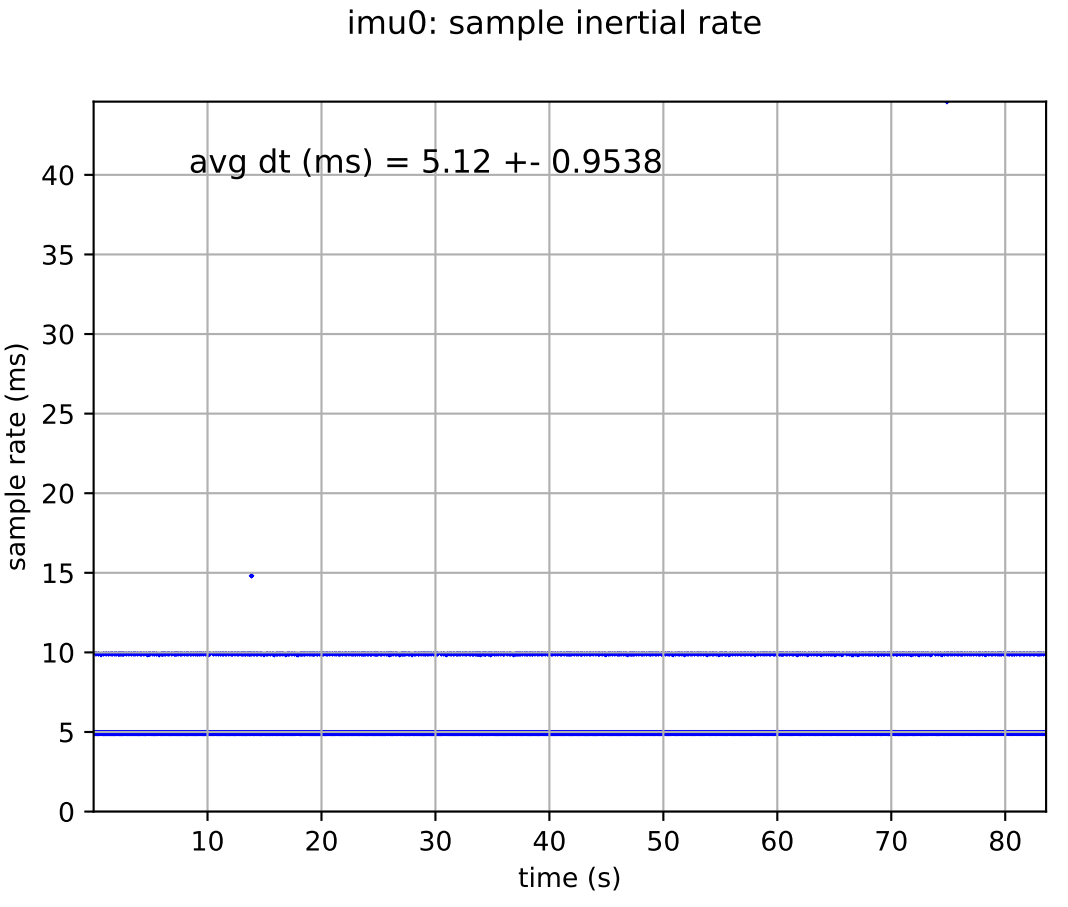
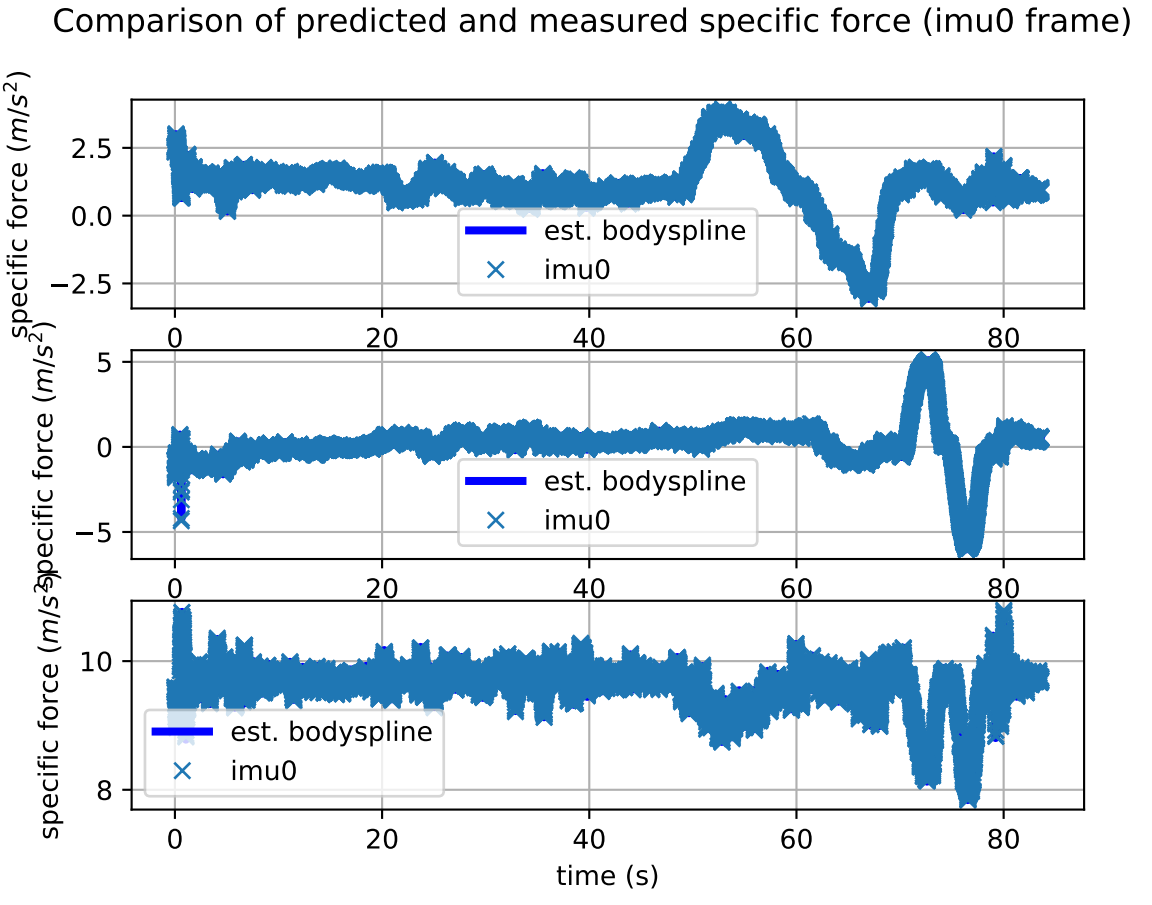
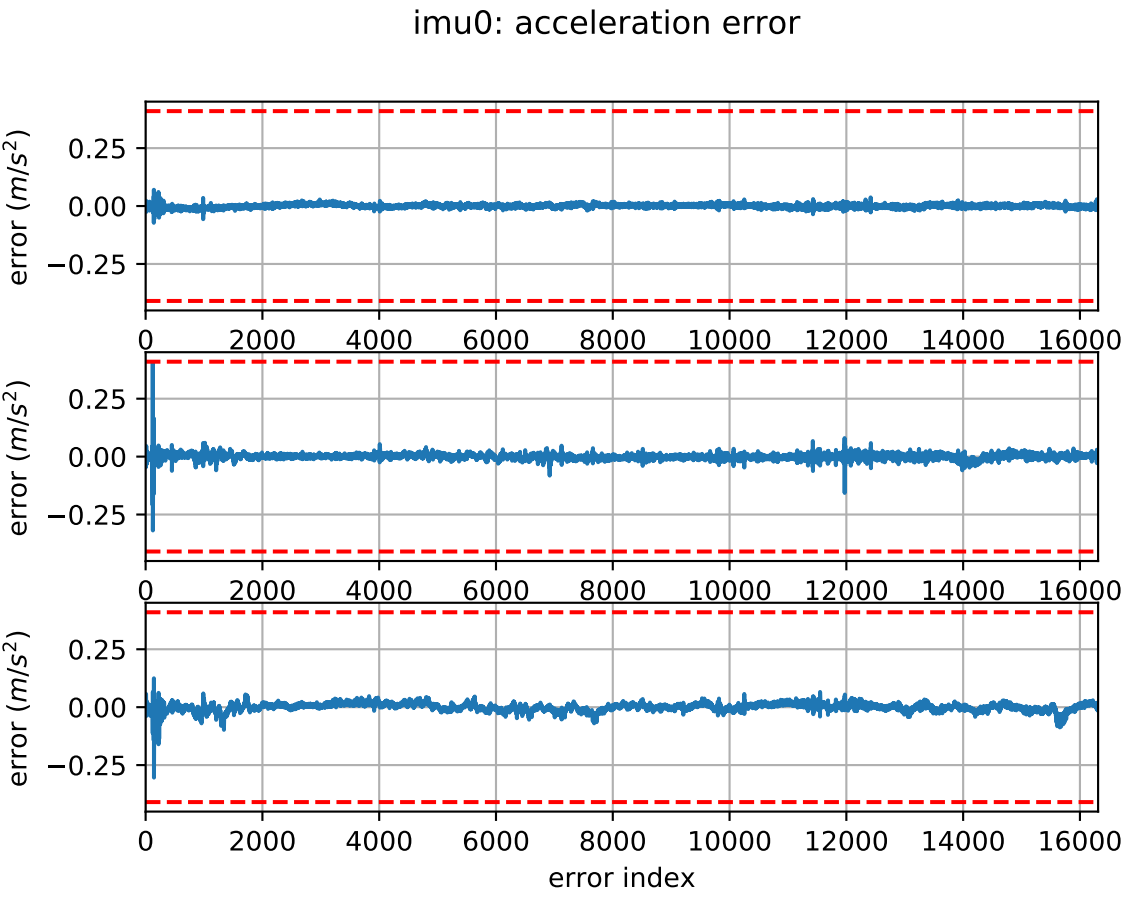
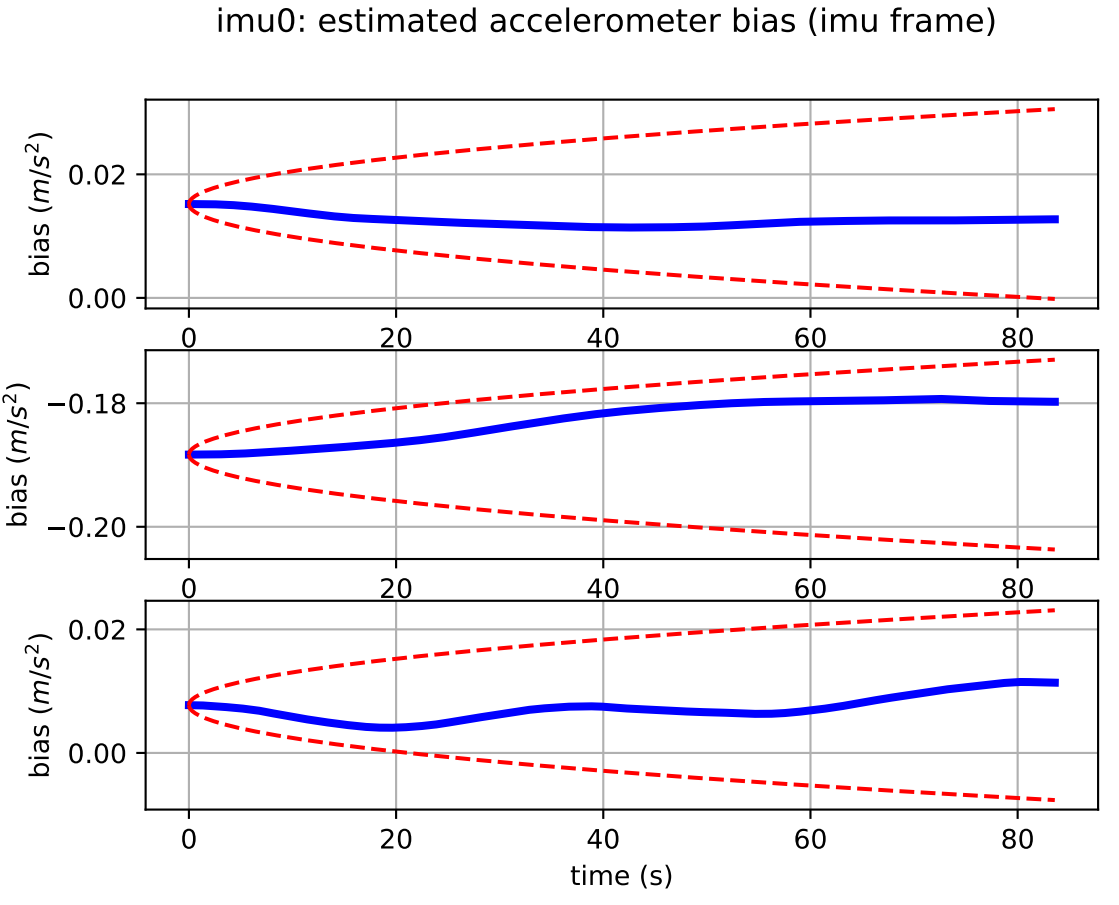

imu0: estimated poses



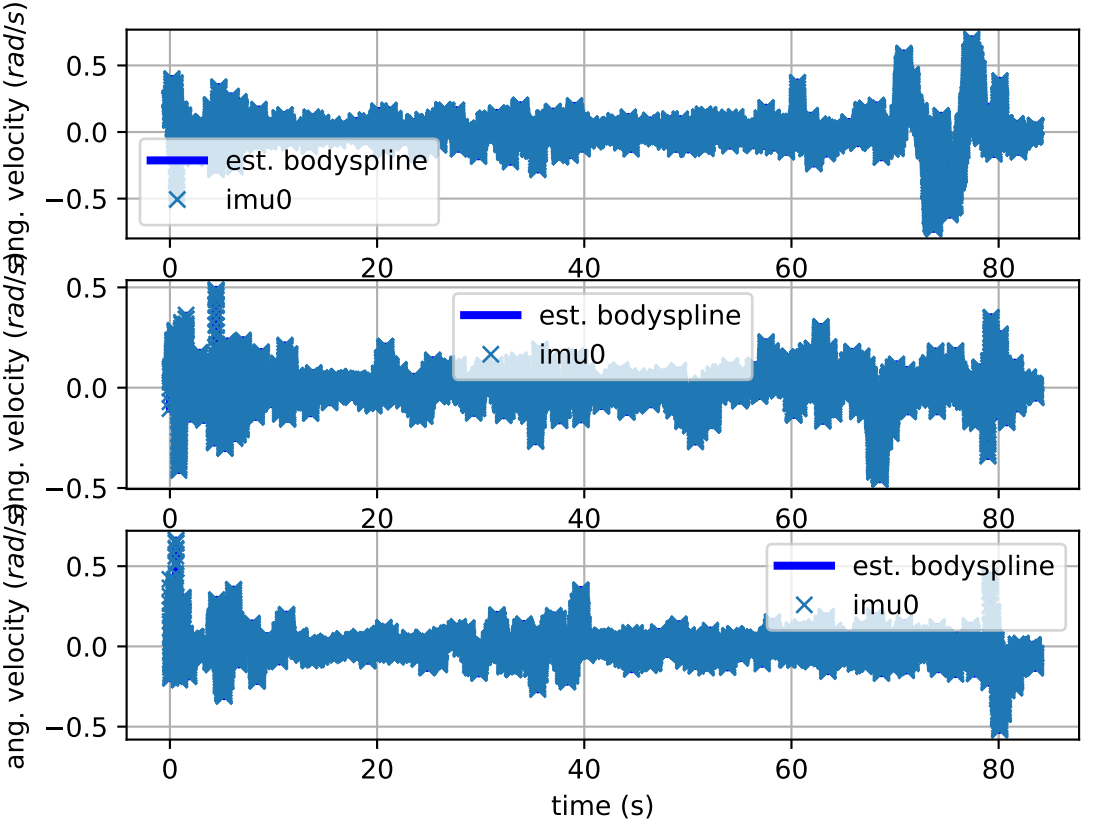

imu0: sample inertial rate
avg dt (ms) = 5.12 +- 0.9538
sample rate (ms)
0
5
10
15
20
25
30
35
40
time (s)
10
20
30
40
50
60
70
80

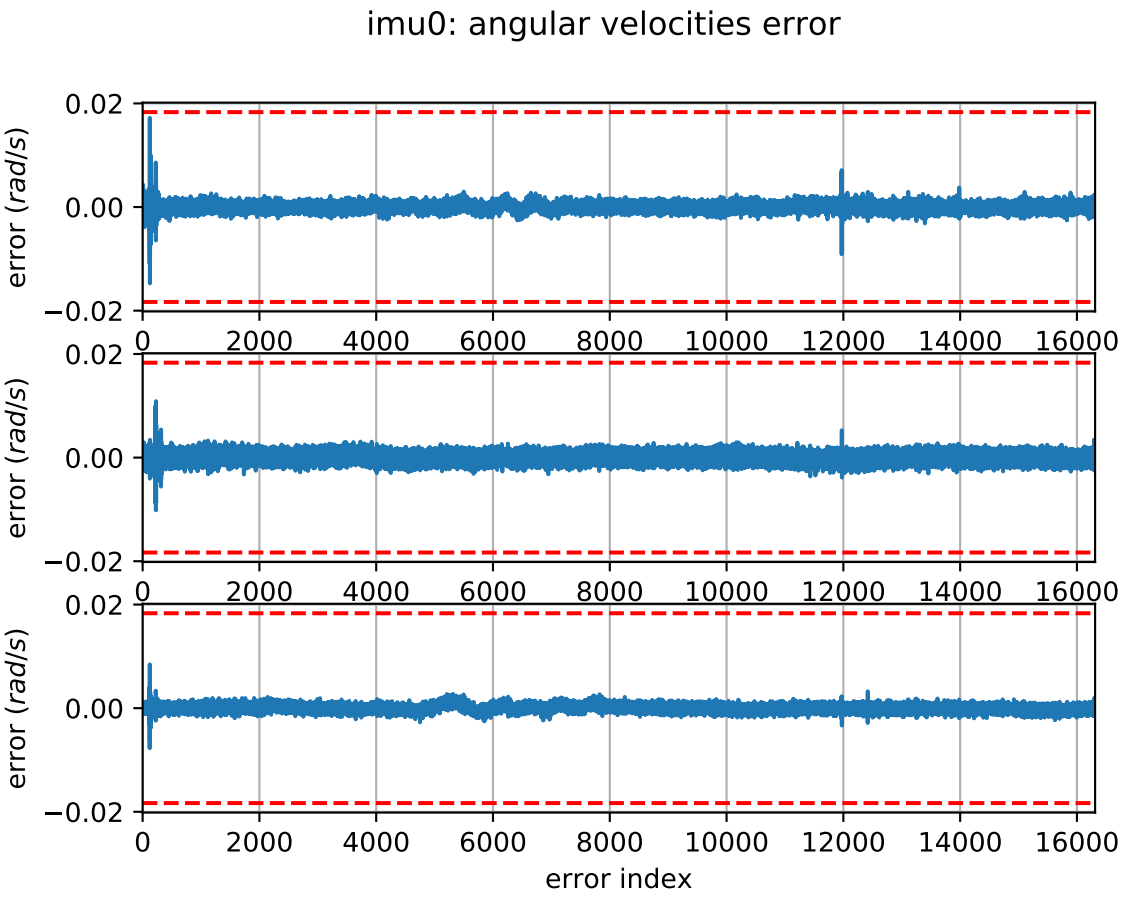

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
2.5
0.0
−2.5
est. bodyspline
imu0
0
20
40
60
80
specific force (m/s^2)
5
0
−5
est. bodyspline
imu0
0
20
40
60
80
specific force (m/s^2)
10
8
est. bodyspline
imu0
0
20
40
60
80
time (s)

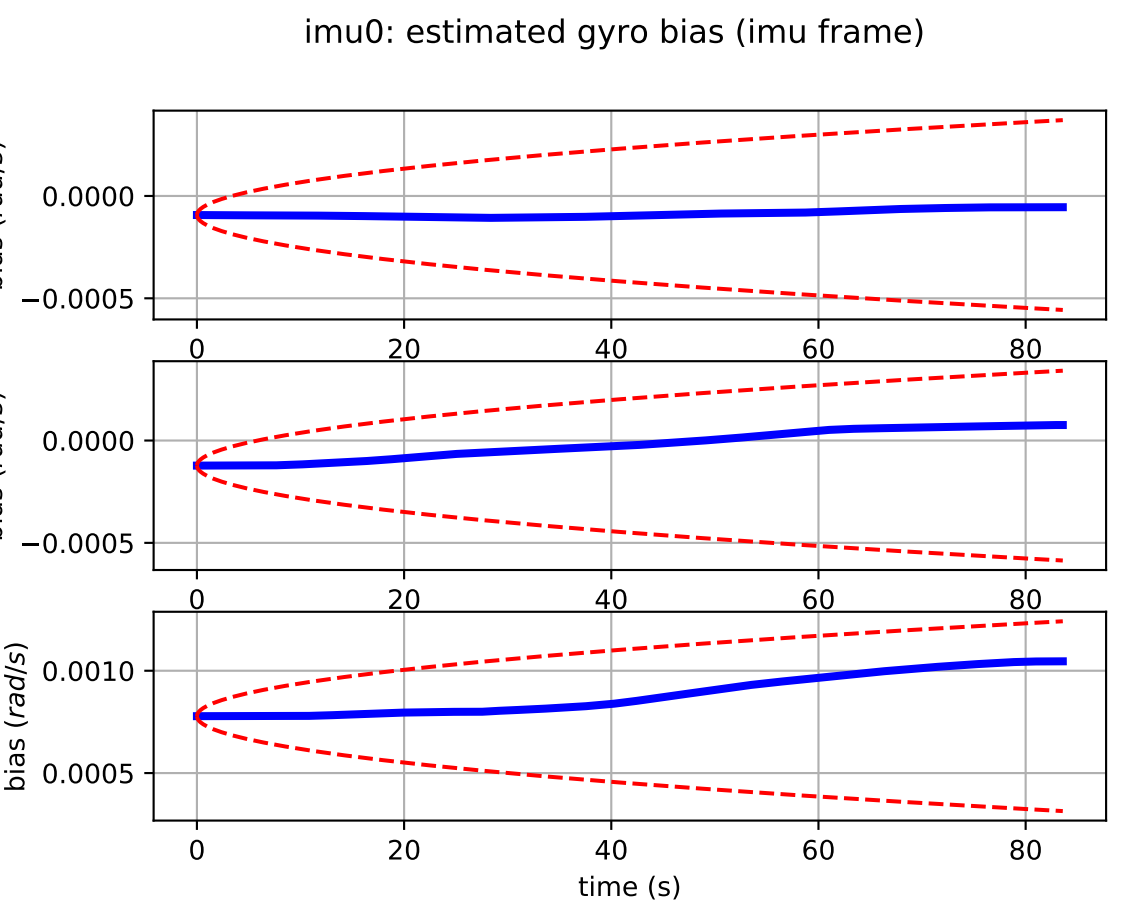

imu0: acceleration error
error (m/s^2)
0.25
0.00
−0.25
0
2000
4000
6000
8000
10000
12000
14000
16000
error index

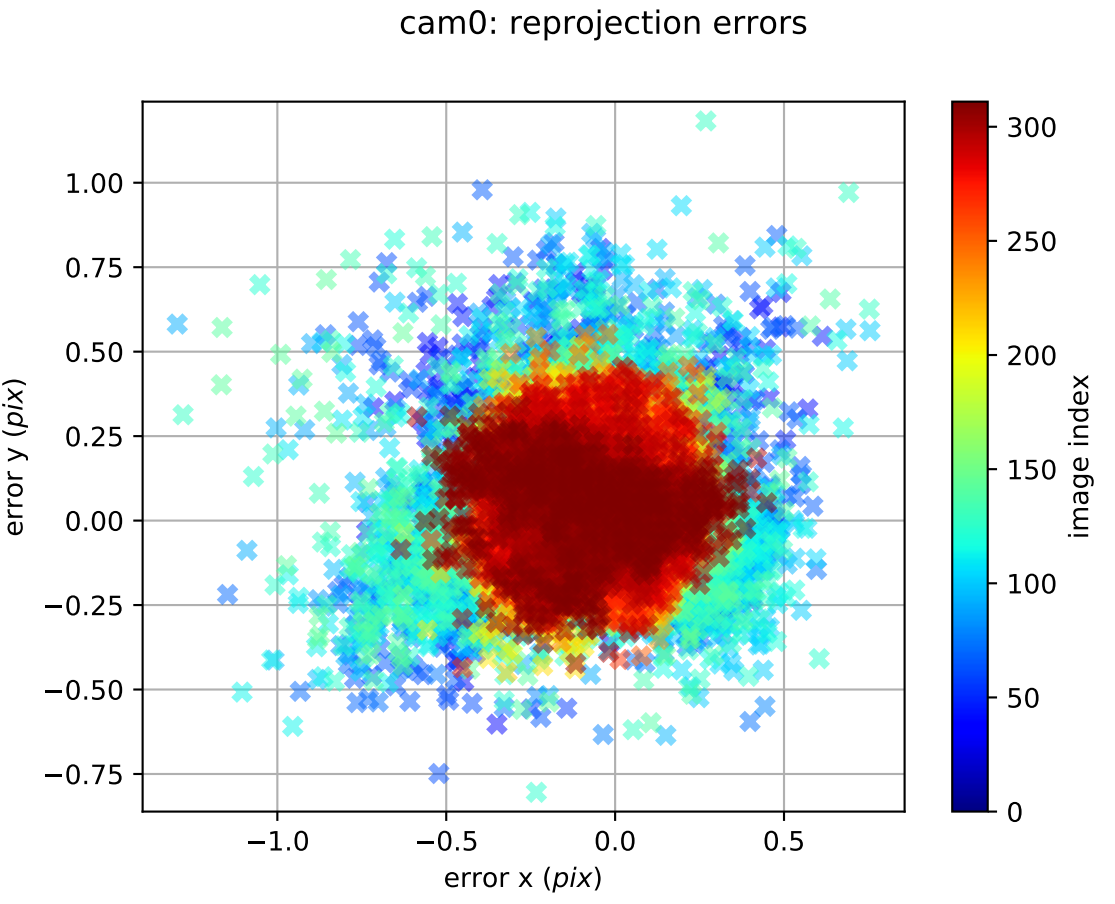

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.02
0.00
0
20
40
60
80
bias (m/s^2)
−0.18
−0.20
0
20
40
60
80
bias (m/s^2)
0.02
0.00
0
20
40
60
80
time (s)

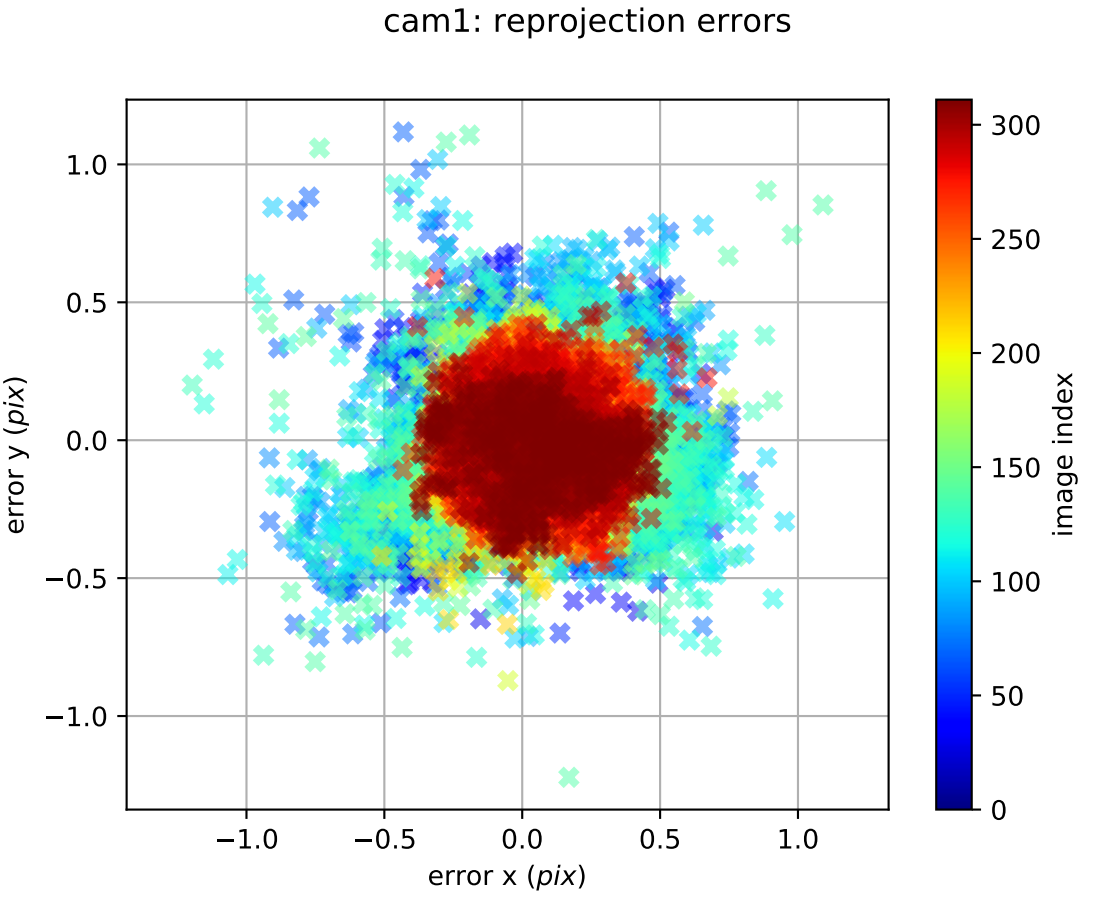
Comparison of predicted and measured angular velocities (body frame)




imu0: angular velocities error
error (rad/s)
0.02
0.00
−0.02
0
2000
4000
6000
8000
10000
12000
14000
16000
error index


imu0: estimated gyro bias (imu frame)
0.0000
−0.0005
0.0000
−0.0005
0.0010
0.0005
bias (rad/s)
0
20
40
60
80
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.0
−0.5
0.0
0.5
300
250
200
150
100
50
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
300
250
200
150
100
50
0